Braillix - Electronics BOM (HAVE on top, BUY at bottom)

> Written 2026-07-30. Companion to `docs/MASTER_BOM.md` (which also covers mechanical parts). Scope: **one working single-cell demo.** Multi-cell/production parts are listed separately at the end so they are not confused with what you need now.

? The headline: you need NO chips, NO resistors, NO capacitors

Every discrete component the circuit would normally need is **already built into the modules you own**:

What you'd normally add	Why you don't need it
Motor flyback diodes	**Built into the ULN2003A.** The chip has internal freewheeling diodes on its COM pin - that is what COM is for.
Motor driver transistors	**That IS the ULN2003.** Seven Darlington pairs, 500mA each. You need four.
Button pull-up resistors	ESP32 has **internal** pull-ups. Firmware uses `INPUT_PULLUP`.
Hall sensor pull-up / comparator	**On the blue MH module already** (that's the second IC and the trimmer pot).
Voltage regulator + decoupling caps	**On the ESP32 DevKit already** (AMS1117 + caps).
Reverse-polarity protection	[!] **NOT present.** This is why the multimeter matters - see below.

Total passives to buy for the demo: zero.

PART 1 - WHAT YOU ALREADY HAVE

Cross-check each of these physically. The "how to identify it" column is so you can confirm you actually have the right thing.

#	Component	How to identify it	Qty	Status
1	**ESP32 DevKit (USB-C)**	Black PCB ~51x28mm, silver square shield, USB-C at one end, 15 pins per side, BOOT + EN buttons	1	[OK] HAVE (?350)
2	**28BYJ-48 stepper motor**	Silver can ~28mm dia, blue plastic base, 5 coloured wires into a white plug	1	[OK] HAVE (?160 w/ driver)
3	**ULN2003 driver board**	Small **blue** PCB, 4 red LEDs, white 5-pin socket, black 16-pin chip, IN1-IN4 + +/- pins	1	[OK] HAVE - **this is your motor driver, no chip to buy**
4	**Hall sensor module (MH-Sensor-Series)**	Small **blue** PCB, tiny black 3-pin sensor on one edge, blue trimmer pot, 1-2 LEDs, pins marked **AO DO GND VCC**	1	[OK] HAVE (?50) - [!] **must use AO (analog)**, and the pocket won't fit it (see note)
5	**5V / 3A power adapter**	Wall plug, barrel connector on the lead, "5V 3A" on the label	1	[OK] HAVE (?160 w/ jack)
6	**Barrel jack (screw terminal)**	Small yellow/black block, barrel socket one end, two screw terminals the other	1	[!!] HAVE but **polarity NEVER verified** - see below
7	**Connecting wire (2m)**	Loose hookup wire	2m	[OK] HAVE (?20)
8	**Dupont jumper wires**	Ribbon of coloured wires with plastic pin housings	some	[OK] HAVE - need ~8x M-F and ~5x M-M
9	**Neodymium magnets 8x1mm**	Small silver discs, strongly attract each other	10	[OK] HAVE (?135) - for docking, **not** for cam homing

#	Component	How to identify it	Qty	Status
10	**Tactile switches 6x6x5mm**	Tiny square black buttons, 4 legs	3	[OK] HAVE - not needed for the demo
11	**Soldering iron**	-	1	[OK] HAVE

[!!] Two warnings about things you already have

#6 - the barrel jack polarity is still unverified. There is no reverse-polarity protection anywhere in this circuit. If the centre pin is negative and you wire it as positive, the ESP32 dies instantly and silently. **This is the single most likely way to lose ?350 in this project.** Check it with a multimeter before anything is powered. Procedure is in `ASSEMBLY_BIBLE.md` Step 1.

#4 - your hall module does not fit its pocket. The base plate pocket is 5.3 x 4.3 x 3mm, designed for a *bare* TO-92 sensor, not a PCB module. (It also turns out that pocket is entirely swallowed by the cam pocket, so it doesn't exist on the printed part at all - see `CAD_FIT_CHECK.md`.) **For the demo this does not matter** - the module sits on the breadboard. For assembly later you either desolder the bare 3-pin sensor off the module, or redesign the pocket.

PART 2 - WHAT YOU NEED TO BUY

2A. For the demo (buy this week)

#	Component	Spec	Why	Est. ?
1	**Digital multimeter**	Any basic DMM with DC volts	[!!] **MANDATORY.** Verifies barrel jack polarity before power-up. Also your only way to debug a dead rail.	~500
2	**Breadboard**	Half-size or full-size, 400-830 tie points	The whole Tier-0 demo is breadboard-based, and it's the safe way to bring up wiring before soldering.	~100
3	**USB-C cable (DATA)**	Must carry data, not charge-only	To flash the ESP32. Charge-only cables look identical and fail silently. **Test the one you have first - you may already be fine.**	~150
4	**Solder wire**	**0.8mm, 60/40 rosin core**	For soldering wires flat to the ULN2003. Not plumbing solder.	~150
5	**Wire stripper / cutter**	Small, for 22-26 AWG	Prepping wire ends.	~200

Demo subtotal: ~?1,100 (or ~?950 if your existing USB-C cable does data)

2B. For the mechanism (buy when the resin parts arrive)

#	Component	Spec	Qty	Est. ?
6	**Micro compression springs**	**2.0mm OD**, ~0.3mm stainless wire, ~4mm free length. Buy an assortment kit. [!!] Pen springs (4mm) **cannot** work - braille rows are 2.6mm apart.	6 + spares	~500
7	**Homing magnet**	**3mm dia x 2mm thick** neodymium. Your 8x1mm ones will **not** fit the cam pocket.	1 (buy 10)	~120
8	**M2.5 x 25mm bolts**	Top plate -> standoffs -> box bosses	4	~60
9	**M4 x 10mm bolts + nuts**	Motor mounting ears	2	~30

#	Component	Spec	Qty	Est. ?
10	**M2 x 8mm self-tap**	Pod lid. **NOT M2x6** - through a 4mm lid that leaves only 2mm of thread.	2	~40
11	**Superglue or 5-min epoxy**	Glues the resin dot insert into the PETG plate. Epoxy preferred - you get time to seat it square.	1	~80
12	**Digital calipers**	150mm digital	[!!] Both open project blockers are *measurements* (motor x8, jack body).	1

Mechanism subtotal: ~?1,430

2C. Optional / nice to have

Component	Why	Est. ?
Flux paste	Makes soldering to ULN2003 pins much easier for a beginner	~100
Heat-shrink tubing	Insulating soldered joints	~80
Tweezers	Handling 2mm springs and 1mm linkages - genuinely hard without	~80
Desoldering pump/wick	Fixing bridges; also for lifting the TO-92 off the hall module	~120
Small drill bits 1.5/1.7/2.5mm	FDM holes print 0.2-0.3mm undersize	~150
Electrical tape	-	~30

Totals

	?
Already spent	935
Demo essentials (2A)	~1,100
Mechanism (2B)	~1,430
Optional (2C)	~560
Everything	**~4,025 spent, of ?15,000**

If you buy only 2A (~?1,100) you can run the full presentation demo.

PART 3 - NOT NEEDED (so you don't buy it by mistake)

Item	Verdict
Any motor driver IC	[X] You own it. The chip on the blue board **is** the ULN2003A.
ATmega328P / custom muscle board	[X] Only for multi-cell I2C chains. The ESP32 drives one cell directly. Also TQFP-32 at 0.8mm pitch - beyond basic soldering; would need assembled (PCBA) ordering.
Resistors	[X] None for a single cell. *(Only if you later build the multi-cell I2C bus: 2x 4.7k? pull-ups, master end only.)*
Capacitors	[X] None. Both modules and the DevKit carry their own.
Flyback diodes	[X] Built into the ULN2003A (that's what COM does).
Level shifters	[X] Not needed - hall runs on 3V3, so GPIO34 never sees 5V.
2mm bearing balls	[X] Obsolete since v7.1 - the dot is printed as a dome.
Pogo pin connectors	?? Deferred. Spec when needed: 4-pin, 2.54mm pitch, spring-loaded, ~0.5A.

Item	Verdict
Crystal / oscillator	[X] ESP32 and the modules have their own.

? If you ever DO build the custom muscle board

Not needed now, but for the record - it's `pcb/braillix_muscle_board.kicad_pcb`, 34x44mm:

Part	Package	Note
ATmega328P-AU	**TQFP-32, 7x7mm square**	0.8mm pitch - **not hand-solderable at your level**
ULN2003A	**SOIC-16 rectangle**	Same chip you already own, just SMD
16MHz crystal	HC49	
2x 22pF, 2x 100nF	0402	Tiny - 1.0 x 0.5mm
100µF/10V	SMD electrolytic	
10k? x2, 4.7k? x2	0402	
SS14 Schottky	SMA	?? *this* is the reverse-polarity protection the prototype lacks

Recommendation: order it fully assembled (PCBA) or don't build it. 0402 passives and a TQFP-32 are not a beginner job, and none of it is needed for a working demo.